

Nandana Dileep

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PROFILE

AI Engineer building production agentic systems, LLM infrastructure, and developer tools. At Jaguar Land Rover, developing an AI-native spec-driven engineering platform for software-defined vehicles. Creator of Mycelium, an open-source runtime safety layer for AI agents.

PROFESSIONAL EXPERIENCE

AI Engineer (GSET), *Jaguar Land Rover, SDV*, Bangalore, India July 2024 - Present

- **Sarvam (Spec-Driven Development)**: Core developer on an 8-person SDV team building an AI-native Everything-as-Code platform for automotive requirements, HLD/LLD, tests, and compliance. Built around version-controlled AsciiDoc, GitLab CI/CD quality gates, AWS Bedrock-assisted authoring and review, and a knowledge-graph digital thread; adopted by a major vehicle program.
- **Production ML Optimization**: Deployed SMOTE-balanced ensemble classifier identifying underutilized test rigs, achieving **18% cost reduction** through asset consolidation (processed 500K+ telemetry events daily).
- **Deep Learning Pipeline**: Developing core components of a multi-stage validation system combining heuristic algorithms and Deep Learning to rank vehicle test cases; integrated JIRA APIs for automated dynamic prioritization.
- **GNSS Localization**: Engineered a C++ GNSS processing pipeline using the GTSAM factor graph library, significantly improving vehicle coordinate estimation accuracy. Presented original research at JLR Tech Symposium.

Software Development Engineer Intern, *Amazon*, Bangalore, India May 2023 - July 2023

- **Distributed Systems**: Designed and deployed a Latency Analysis Service for Alexa using AWS Lambda, Coral, and OpenSearch; completed production deployment ahead of schedule.
- **Test Automation**: Built end-to-end testing framework using JUnit 5 and Mockito, increasing code reliability and reducing deployment rollback rates.

RESEARCH & PROJECTS

Mycelium: Runtime Safety for AI Agents, *In Development* Docs [🔗](#), GitHub [🔗](#), PyPI [🔗](#)

- Building an open-source runtime that prevents predictable agent failures before tool execution, including duplicate side effects on retry, stale context, invalid inputs, and out-of-scope tool calls.
- Published the framework-independent `mycelium-runtime` Python SDK on PyPI for LangGraph, CrewAI, and custom agent loops; durable transition keys safely resolve repeated reads and block ambiguous writes.

KAIROS / Auto-Agent: Autonomous Background Coding Agent GitHub [🔗](#)

- Built an autonomous daemon that gathers context from git history, TODOs, modified files, stale pull requests, and environment configuration, then uses an LLM scheduler to decide whether to act or sleep.
- Added presence-aware notifications and nightly “AutoDream” memory consolidation, compressing observations into persistent project memory while pruning contradictions.

vLLM-Inspired LLM Serving Engine on Apple Silicon GitHub [🔗](#)

- Built an MLX-based serving engine from scratch with PagedAttention-inspired KV cache management, continuous batching, chunked prefill, prefix caching, and asynchronous request scheduling.
- Exposed OpenAI-compatible chat completions and benchmarked the implementation against vLLM and llama.cpp; published an architectural deep dive.

Identiti: Personal AI Memory Layer with Knowledge Graphs Live [🔗](#), GitHub [🔗](#)

- Built a production full-stack memory system that extracts confidence-scored identity entities from conversations and stores their relationships in a Neo4j knowledge graph.
- Implemented React/Vite, Flask, Redis, Supabase authentication, per-user field-level encryption, and portable “Memory Card” export; deployed on Vercel and Render.

WRITING

“[Checklist to Prevent All the Agent Failure Modes](#)” [🔗](#) Medium, 2026
Systematic framework for diagnosing and preventing failure modes in agentic LLM systems.

- [“Building a vLLM-Inspired LLM Serving Engine on Apple Silicon with MLX”](#)

Medium, 2026
- Architectural deep dive into production LLM serving: PagedAttention-inspired KV cache management, continuous batching, and async scheduling built from scratch on Apple Silicon.
- [“Swapped Every Part of nanoGPT With Modern Alternatives”](#)

Medium, 2026
- Component-by-component ablation study of modern transformer techniques on nanoGPT; benchmarked each substitution independently and published results.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL

Agentic AI: Tool Calling, MCP, Agent Memory, Guardrails, Evaluation, RAG, Knowledge Graphs

ML/LLMs: PyTorch, Transformers, PEFT, MLX, vLLM, FAISS, Scikit-learn

Backend/Infrastructure: FastAPI, Flask, Redis, PostgreSQL, Neo4j, Docker, AWS, GitHub Actions, GitLab CI

EDUCATION

Indian Institute of Technology Madras (IIT Madras) - Bachelor of Technology

Aug 2020 - May 2024

Major: Civil Engineering

PUBLICATIONS

Precise Position Estimation using Factor Graph Optimization

Jaguar Land Rover Tech Symposium, June 2024. Original research on GNSS localization using factor graph methods in C++ with GTSAM.